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Battery Management System Equipped with Parking to Prevent Parasitic Leakage

Abstract

Systems and methods for reducing parasitic current leakage. A switch includes a source resistor, a first transistor, a second transistor, and a midpoint transistor. The midpoint transistor is located in between the first transistor and the second transistor. The midpoint transistor is maintained at a low voltage in response to the switch being in a non-conductive state, and the low voltage is less than or equal to an output voltage of the switch.

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Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to battery management systems and specifically relates to systems and methods for preventing leakage in battery management systems.

BACKGROUND

[0002] Battery management systems (BMS) are designed to monitor, protect, and manage the performance of batteries. Battery management systems may specifically be used in electric vehicles, portable electronic devices, renewable energy systems, and so forth.

[0003] Current leakage in battery management systems can be a concern in situations wherein the battery management system is expected to monitor and manage the state of rechargeable batteries. Excessive current leakage can lead to several issues, including reduced battery life, inefficient energy usage, and potential safety hazards. Battery management systems are typically designed to operate with minimal power consumption, especially in low-power modes when the device is not actively using the battery. However, some low-level current leakage is common due to the intrinsic properties of components of the battery management system, and further due to power requirements.

[0004] In some cases, a battery management system includes a parasitic BJT (bipolar junction transistor). High-voltage NMOS (N-channel metal-oxide semiconductor) transistors typically have a drain connected to an isolating N-iso well. In some cases, there is a parasitic PNP (positive-negative-positive) transistor formed in the body of the NMOS transistor. This parasitic PNP transistor can cause current leakage that reduces performance of the battery management system and the connected batteries.

[0005] Considering the foregoing, disclosed herein are systems, methods, and devices for improved battery management that prevent current leakage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Non-limiting and non-exhaustive implementations of the disclosure are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified. Advantages of the disclosure will become better understood regarding the following description and accompanying drawings where:

[0007] FIG. 1 is a schematic block diagram illustrating some components of a system for battery management;

[0008] FIG. 2A is a schematic circuit diagram illustrating a simplified portion of a multiplexer with three switches;

[0009] FIG. 2B is a schematic circuit diagram illustrating a simplified portion of a multiplexer with three switches;

[0010] FIG. 3 is a schematic circuit diagram of a multiplexer that includes three switches, wherein each of the three switches includes dual series MOSFET implementation;

[0011] FIG. 4 is a schematic circuit diagram of a multiplexer that includes three switches, wherein each of the three switches includes dual series MOSFET implementation;

[0012] FIG. 5 is a schematic cross-sectional diagram of an NMOS transistor that includes a drain connected to an isolating N-iso well;

[0013] FIG. 6 is a schematic circuit diagram of a switch that includes a dual series MOSFET implementation with a parasitic PNP (positive-negative-positive) transistor;

[0014] FIG. 7 is a schematic circuit diagram of a switch that includes a dual series MOSFET implementation with a parasitic PNP (positive-negative-positive) transistor;

[0015] FIG. 8 is a schematic circuit diagram of a switch that includes a plurality of transistors configured to implement transistor voltage parking to reduce current leakage;

[0016] FIG. 9 is a schematic circuit diagram of a switch that includes a plurality of transistors configured to implement transistor voltage parking to reduce current leakage; and

[0017] FIG. 10 is a schematic circuit diagram of a switch that includes a plurality of transistors

configured to implement transistor voltage parking to reduce current leakage.

DETAILED DESCRIPTION

[0018] Described herein are systems, methods, and devices for battery management. The battery management systems and devices described herein are configured to measure a stack of batteries and connect battery inputs into one or more analog-to-digital converters (ADCs). Specifically described herein are battery management systems and devices capable of executing a parking function to reduce current leakage.

[0019] The battery management systems described herein may be implemented with asymmetric high-voltage MOSFETs (metal oxide semiconductor field-effect transistors). High-voltage MOSFETs are a type of power electronic device designed to handle high voltage levels in various applications. MOSFETs are used in electronic circuits to control the flow of current, and high-voltage MOSFETs are specifically engineered to operate at elevated voltage levels. High-voltage MOSFETs are optimized for low on-resistance, which helps minimize power losses and heat generation when the transistor is conducting current. Lower on-resistance values can be important for efficient high-voltage applications. High-voltage MOSFETs are useful in various fields, including power electronics, automotive systems (e.g., electric vehicles and battery management), renewable energy systems (e.g., solar inverters), and industrial control systems.

[0020] The battery management systems described herein may further be implemented with a high-voltage DMOS (double diffused metal oxide semiconductor). High-voltage DMOS transistors are a type of power semiconductor that combines the advantages of DMOS technology with the capability to handle high voltage levels. High-voltage DMOS may be implemented in various applications where high-voltage switching, or amplification is required. High-voltage DMOS devices are designed to operate at elevated voltage levels and are optimized for low on-resistance. This helps minimize power losses and improves efficiency in high-voltage applications. High-voltage DMOS transistors often require a higher gate-source voltage to turn on when compared to lower voltage MOSFETs. Adequate gate driver circuits may be employed to provide the necessary voltage and current for proper switching.

[0021] The battery management systems described herein may further be implemented with an NMOS (N-channel metal-oxide semiconductor) transistor. The NMOS transistor is a type of field-effect transistor (FET) used in electronic circuits, and especially in digital integrated circuits. NMOS transistors are capable of controlling the flow of current between two terminals, which are referred to as the source terminal and the drain terminal. The NMOS utilizes a third terminal, referred to as the gate terminal, to control the flow of current between the source terminal and the drain terminal. NMOS transistors apply a voltage to the gate terminal, and when a positive voltage is applied to the gate terminal relative to the source terminal, the NMOS transistor creates an electric field in the channel region beneath the gate terminal. This electric field attracts electrons from the source terminal to form a conductive channel between the source terminal and the drain terminal. NMOS transistors are normally off (i.e., non-conductive) until a gate-source voltage is applied to turn them on. The gate source voltage is typically positive relative to the source terminal to create the electric field needed for conduction. When an NMOS transistor is in the on state (i.e., conductive), electrons flow from the source terminal to the drain terminal. In digital circuits, NMOS transistors are often utilized to implement logic states. By connecting multiple NMOS transistors in various configurations, logic functions can be created.

[0022] For the purposes of promoting an understanding of the principles in accordance with the disclosure, reference will now be made to the embodiments illustrated in the drawings and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the disclosure is thereby intended. Any alterations and further modifications of the inventive features illustrated herein, and any additional applications of the principles of the disclosure as illustrated herein, which would normally occur to one skilled in the relevant art and having possession of this disclosure, are to be considered within the scope of the disclosure

claimed.

[0023] Before the structure, systems, and methods described herein are disclosed and described, it is to be understood that this disclosure is not limited to the particular structures, configurations, process steps, and materials disclosed herein as such structures, configurations, process steps, and materials may vary somewhat. It is also to be understood that the terminology employed herein is used for the purpose of describing particular embodiments only and is not intended to be limiting since the scope of the disclosure will be limited only by the appended claims and equivalents thereof.

[0024] In describing and claiming the subject matter of the disclosure, the following terminology will be used in accordance with the definitions set out below.

[0025] It must be noted that, as used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise.

[0026] As used herein, the terms “comprising,” “including,” “containing,” “characterized by,” and grammatical equivalents thereof are inclusive or open-ended terms that do not exclude additional, unrecited elements or method steps.

[0027] As used herein, the phrase “consisting of” and grammatical equivalents thereof exclude any element or step not specified in the claim.

[0028] As used herein, the phrase “consisting essentially of” and grammatical equivalents thereof limit the scope of a claim to the specified materials or steps and those that do not materially affect the basic and novel characteristic or characteristics of the claimed disclosure.

[0029] Referring now to the figures, FIG. 1 is a schematic block diagram illustrating some components of a system **100** for battery management. The system **100** specifically illustrates a multiplexer (may be abbreviated “MUX” as described herein), which is a digital electronic device, analog device, or a combinational logic circuit that serves as a data selector. The system **100** allows multiple input data lines to be routed to a single output line. The system **100** may be implemented in a digital electronics device and may be utilized for data routing, signal selection, and control.

[0030] The system **100** includes a multiplexer mode module **102** and a timing control module **104**, which may collectively be referred to herein as the control modules **102**, **104**. The system **100** further includes a positive charge pump **106** and a negative charge pump **114**. The control modules **102**, **104** are in electronic communication with each of the positive charge pump **106** and the negative charge pump **114**.

[0031] The system **100** includes three sets of high-voltage multiplexers and current switch controls. Specifically, the system **100** includes a first current switch control **110a** in electrical communication with a first high-voltage multiplexer **108c**. The system **100** includes a second current switch control **110b** in electrical communication with a second high-voltage multiplexer **108b**. The system **100** includes a third current switch control **110c** in electrical communication with a third high-voltage multiplexer **108c**. The first high-voltage multiplexer **108c** outputs a first multiplexer output **112a**; the second high-voltage multiplexer **108b** outputs a second multiplexer output **112b**; and the third high-voltage multiplexer **108c** outputs a third multiplexer output **112c**. The system **100** additionally includes a matched current generator **116** that is in electrical communication with each of the first current switch control **110a**, the second current switch control **110b**, and the third current switch control **110c**.

[0032] The system **100** may be altered depending on the desired implementation. Specifically, the quantity of high-voltage multiplexers **108a-108c** may be adjusted depending on the desired implementation. In a specific use-case described herein, the system **100** is utilized as a component of a battery management system that controls numerous battery cell inputs. However, the components of the system **100** and the architecture of the system **100** may be implemented in other use-cases not involving battery management.

[0033] The example system **100** illustrated in FIG. 1 may be implemented in a battery management system that includes eighteen battery cell inputs, or another quantity of battery cell inputs that is

divisible by the quantity of high-voltage multiplexers **108a-108c**. In an alternative implementation, system **100** is implemented in a battery management system, and the quantity of battery cell inputs is not divisible by the quantity of high-voltage multiplexers **108a-108c**. Depending on the use-case, it is not required that each high-voltage multiplexer **108a-108c** has the same number of inputs.

[0034] The system **100** is configured to take the battery cell inputs, general purpose input/output (GPIO) signals, and sensor signals, and then multiplex those signals to three different analog-to-digital converters (ADCs). The quantity of ADCs may correspond with the quantity of high-voltage multiplexers **108a-108c**, and thus, the quantity of ADCs may be adjusted if the system **100** is altered to include a different number of high-voltage multiplexers **108a-108c** than shown in the example of FIG. 1.

[0035] The system **100** is divided across the three high-voltage multiplexers **108a-108c**, which select from the battery cells in groups of six. Thus, each of the three high-voltage multiplexers **108a-108c** is assigned to six of the 18 battery cells. The high-voltage multiplexers **108a-108c** connect the battery cells to three ADCs. In some implementations, the high-voltage multiplexers **108a-108c** may additionally connect to a low-voltage multiplexer (not shown in FIG. 1) that is configured to select from the GPIO signals and various sensor signals. Each of the multiplexers in the system **100** is differential. The high-voltage multiplexers **108a-108c** switch the top and bottom of the selected battery cells to their outputs. The auxiliary multiplexer switches a GPIO signal or a sensor signal to its positive output, the ground (if a GPIO is selected), or the appropriate ground sense (if a sensor is selected).

[0036] The switches (not shown in FIG. 1) float the various battery cell inputs but may be configured to tolerate only 6 V maximum across their gate oxides. The switches may be turned on by sending a current through a resistor connected from gate to source. At the same time, an identical current may be sunk from the source of the switch, such that the net current flowing through the switch and back to the battery cell input is minimized to the extent possible given the accuracy of the matching of the source and sink currents.

[0037] The currents passing through the system **100** are generated by an amplifier that forces a fraction of a regulated 5 Volt input across a replica resistor. Multiple matched copies of the current through that resistor may be mirrored for the source and sink currents of the enabled switches. The gate/source resistors are the same type and may have the same width and segment length as the replica resistor. This ensures the process and temperature variation of the resistance is nominally cancelled, and the gate-source voltage (V_{gs}) of the switches is a function of the regulated 5 Volt input.

[0038] The current switch controls **110a-110c** are configured to switch the currents from the matched current source array to the enabled switches. In some cases, the system **100** may be controlled with a 5 V logic block that generates all individual controls for each of the high-voltage multiplexers **108a-108c** and current sources.

[0039] The system **100** includes the positive charge pump **106** and the negative charge pump **114**. The negative charge pump **114** may specifically include a negative inverting doubler charge pump that is configured to create a nominal -5 V to serve as the negative rail for the current sources that sink current from the lowest switch, whose channel voltage may be at or near ground. The positive charge pump **106** may specifically include a positive tripler charge pump that is configured to create a nominal positive supply voltage (" V_{DD} ") $+10$ V to serve as the positive rail for the current sources that source current for a subset of switches whose channels are at or near the positive supply voltage. In an alternative implementation, the positive charge pump **106** includes a positive doubler charge pump.

[0040] FIGS. 2A and 2B are schematic circuit diagrams each illustrating a simplified portion of a multiplexer with three inputs. Each of the multiplexers includes three inputs, including a 40 V input, a 30 V input, and a 20 V input. The multiplexers illustrated in FIGS. 2A and 2B are implemented with different configurations.

[0041] In FIG. 2A, a top switch of the multiplexer is switched into the “on” position such that the multiplexer outputs 40 V. In FIG. 2B, the bottom switch of the multiplexer is switched into the “on” position such that the multiplexer outputs 20 V. In both cases, the center switch, which is connected to a 30 V input, is in the “off” position. In FIG. 2A, there is a negative voltage across the center switch from the input to the output. By contrast, in FIG. 2B, there is a positive voltage across the center switch from the input to the output. Because a single high-voltage transistor can only withstand a large positive voltage in one direction, each switch is made of two series MOSFETs in opposite directions. In this implementation, a MOSFET withstands the voltage in the configuration of FIG. 2A, and another MOSFET withstands the voltage in the configuration of FIG. 2B.

[0042] The switch configurations illustrated in FIGS. 2A and 2B may be implemented in the system **100** illustrated in FIG. 1. The multiplexer illustrated in FIG. 2A includes three switches, including a top switch with a 40 V input, a center switch with a 30 V input, and a bottom switch with the 20 V input. Each of these three switches has the same 40 V output. The multiplexer illustrated in FIG. 2B similarly includes three switches, including a top switch with a 40 V input, a center switch with a 30 V input, and a bottom switch with a 20 V input. Each of these three switches has the same 20 V output.

[0043] FIGS. 3 and 4 are schematic circuit diagrams each illustrating variations on the same circuits illustrated in FIGS. 2A and 2B, but with dual NMOS (N-channel metal oxide semiconductor) transistors. Specifically, the multiplexer **300** illustrated in FIG. 3 corresponds with the multiplexer arrangement illustrated in FIG. 2A. Further, the multiplexer **400** illustrated in FIG. 4 corresponds with the multiplexer arrangement illustrated in FIG. 2B.

[0044] FIG. 3 is a schematic circuit diagram of a multiplexer **300** comprising three inputs, including a top switch that inputs 40 V, a center switch that inputs 30 V, and a bottom switch that inputs 20 V. The multiplexer **300** outputs 40 V. The multiplexer **300** includes a dual series NMOS implementation, wherein each input feeds into dual NMOS transistors.

[0045] The multiplexer **300** includes two switches experiencing a high voltage drop. In this case, the high voltage drain-to-source voltage is higher than a maximum allowed gate-to-source voltage. It is not necessary that the multiplexer **300** actually withstands high voltage, even though the multiplexer **300** is designed to withstand a high voltage drop.

[0046] The multiplexer **300** includes a gate associated with each input. The bottom switch (i.e., the 20 V input) is associated with a first gate **302a**. The center switch (i.e., the 30 V input) is associated with a second gate **302b**. The top switch (i.e., the 40 V input) is associated with a third gate **302c**. Additionally, the multiplexer **300** includes a first source resistor **304a** in electrical communication with the first gate **302a**; a second source resistor **304b** in electrical communication with the second gate **302b**; and a third source resistor **304c** in electrical communication with the third gate **302c**.

[0047] The multiplexer **300** includes a pair of NMOS transistors for each input. The bottom switch includes a first left NMOS transistor **306a** and a first right NMOS transistor **308a**. The center switch includes a second left NMOS transistor **306b** and a second right NMOS transistor **308b**. The top switch includes a third left NMOS transistor **306c** and a third right NMOS transistor **308c**.

[0048] As shown in FIG. 3, the third gate **302c** (i.e., the one in electrical communication with the top switch) is switched on such that current flows through the third gate **302c**. Thus, a current flows through the third gate **302c** and the third source resistor **304c**. In the configuration illustrated in FIG. 3, this current flowing through the third gate **302c** and the third source resistor **304c** has a gate source voltage (V_{gs}) of 5 V. Further as shown in FIG. 3, the currents driving the middle grouping of transistors (i.e., the second left NMOS transistor **306b** and the second right NMOS transistor **308b**) and the bottom grouping of transistors (i.e., the first left NMOS transistor **306a** and the first right NMOS transistor **308a**) are switched off. Thus, the gate source voltages for the middle grouping and the bottom grouping are equal to zero volts.

[0049] Further as shown in FIG. 3, the second left NMOS transistor **306b** includes a second

parasitic drain/body diode **310b**. Additionally, the first left NMOS transistor **306a** includes a first parasitic drain/body diode **310a**. The parasitic drain/body diodes **310b**, **310a** turn on to limit the reverse voltage drop to within one volt. The high voltage is dropped across the drains of the rightmost transistors with proper polarity. Thus, the second parasitic drain/body diode **310b** turns on to limit reverse voltage drop, and the high voltage instead drops across the drain of the second right NMOS transistor **308b**. Likewise, the first parasitic drain/body diode **310a** turns on to limit reverse voltage drop, and the high voltage instead drops across the drain of the first right NMOS transistor **308a**. In FIG. 3, the high voltage drops across the rightmost NMOS transistors are encircled.

[0050] Like the multiplexer **300** illustrated in FIG. 3, FIG. 4 is a schematic circuit diagram of a multiplexer **400** comprising three inputs, including a top switch that inputs 40 V, a center switch that inputs 30 V, and a bottom switch that inputs 20 V. The multiplexer **400** outputs 20 V. The multiplexer **400** includes a dual series NMOS implementation, wherein each input feeds into dual NMOS transistors.

[0051] The multiplexer **400** illustrated in FIG. 4 represents the reverse of the multiplexer **300** illustrated in FIG. 3. Specifically, in FIG. 4, the transistor pair in the bottom grouping is on, and the transistor pairs in the middle grouping and the top grouping are off. The multiplexer **400** switches are configured to withstand a high voltage with the opposite polarity when compared with the multiplexer **300** illustrated in FIG. 3.

[0052] Like the multiplexer **300** illustrated in FIG. 3, the multiplexer **400** includes a gate associated with each input. The bottom switch (i.e., the 20 V input) is associated with a first gate **402a**. The center switch (i.e., the 30 V input) is associated with a second gate **402b**. The top switch (i.e., the 40 V input) is associated with a third gate **402c**. Additionally, the multiplexer **400** includes a first source resistor **404a** in electrical communication with the first gate **402a**; a second source resistor **404b** in electrical communication with the second gate **402b**; and a third source resistor **404c** in electrical communication with the third gate **402c**.

[0053] The multiplexer **400** includes a pair of NMOS transistors for each input. The bottom switch is in electrical communication with a first left NMOS transistor **406a** and a first right NMOS transistor **408a**. The center switch is in electrical communication with a second left NMOS transistor **406b** and a second right NMOS transistor **408b**. The top switch is in electrical communication with a third left NMOS transistor **406c** and a third right NMOS transistor **408c**.

[0054] As shown in FIG. 4, the first gate **402a** (i.e., the one in electrical communication with the bottom switch) is switched on such that current flows through the first gate **402a**. Thus, a current flows through the first gate **402a** and the first source resistor **404a**. In the configuration illustrated in FIG. 4, this current flowing through the third gate **402a** and the third source resistor **404a** has a gate source voltage (V_{gs}) of 5 V. Further as shown in FIG. 4, the currents driving the middle grouping of transistors (i.e., the second left NMOS transistor **406b** and the second right NMOS transistor **408b**) and the top grouping of transistors (i.e., the third left NMOS transistor **406c** and the third right NMOS transistor **408c**) are switched off. Thus, the gate source voltages for the middle grouping and the bottom grouping are equal to zero volts.

[0055] Further as shown in FIG. 4, the second right NMOS transistor **408b** includes a second parasitic drain/body diode **410b**. Additionally, the third right NMOS transistor **408c** includes a third parasitic drain/body diode **410c**. Notably, the location of the parasitic drain/body diodes are reversed in the multiplexer **400** when compared against the multiplexer **300** discussed in connection with FIG. 3. The parasitic drain/body diodes **410b**, **410c** turn on to limit the reverse voltage drop to within one volt. Thus, in contrast with the multiplexer **300** discussed in connection with FIG. 3, the high voltage is dropped across the drains of the leftmost transistors with proper polarity. Thus, the second parasitic drain/body diode **410b** turns on to limit reverse voltage drop, and the high voltage instead drops across the drain of the second left NMOS transistor **406b**. Likewise, the third parasitic drain/body diode **410c** turns on to limit reverse voltage drop, and the

high voltage instead drops across the drain of the third left NMOS transistor **406c**. In FIG. 4, the high voltage drops across the leftmost NMOS transistors are encircled.

[0056] FIGS. 3 and 4 represent multiplexers **300**, **400** with opposite polarity. In FIG. 4, the switches of the multiplexer **400** must withstand a high voltage with the opposite polarity when compared with the multiplexer **300** illustrated in FIG. 3.

[0057] FIG. 5 is a schematic cross-sectional diagram of an NMOS transistor **500** that includes a drain connected to an isolating N-iso well. The high-voltage NMOS transistors have the drain connected to the isolating N-iso well, and there is a parasitic PNP (positive-negative-positive) transistor formed by the body/N-iso/sub, which turns on when the drain/N-iso voltage drops low enough below the source/body voltage. The NMOS transistor **500** illustrated in FIG. 5 may serve as an electronically controlled switch that allows or blocks the flow of current between its source and drain terminal based on voltage applied to its gate terminal.

[0058] The NMOS transistor **500** includes a sub terminal **502**, a source/body terminal **504**, a gate terminal **506**, and a drain/N-iso terminal **508**. The gate terminal **506** controls the flow of current between the source/body terminal **504** and the drain/No-iso terminal **508**. When a voltage is applied to the gate terminal **506** relative to the source/body terminal **504**, this creates an electric field that either allows or restricts the flow of charge carriers in the channel. The source/body terminal **504** is the connection point through which electrons enter the channel. It is typically connected to the lower voltage in an analog circuit, with the source and drain configured at the same voltage. The drain/N-iso terminal **508** is the connection point through which electrons exit the channel. It is typically connected to a higher voltage relative to the source/body terminal **504**. When used as a switch, the source/body terminal **504** and the drain/N-iso terminal **508** may be at the same voltage.

[0059] The NMOS transistor **500** further includes a p-well **510** and a p-sub **514**, and additionally includes an N-iso well **512** disposed in between the p-well **510** and the p-sub **514**. The N-iso well **512** is an isolated region within the silicon substrate, and it is infused with n-type impurities. The n-type N-iso well **512** is isolated from the surrounding p-type substrates (i.e., the p-well **510** and the p-sub **514**). The N-iso well **512** provides electrical isolation between different components, such as n-type and p-type transistors, within the same integrated circuit. This isolation prevents leakage currents and crosstalk between these components.

[0060] When a positive voltage (higher than a certain threshold voltage) is applied to the gate terminal **506** relative to the source/body terminal **504**, an electric field is generated in the channel region beneath the gate terminal **506**. This electric field attracts electrons from the source/body terminal **504** region into the channel, creating a conductive path between the source/body terminal **504** and the drain/N-iso terminal **508**. When the NMOS transistor **500** is in the “on” state (i.e., conducting), current can flow from the source/body terminal **504** to the drain/N-iso terminal **508**. This allows for the desired signal processing, switching, or amplification. When the gate terminal **506** voltage and the source/body terminal **504** voltage are below the threshold voltage, the NMOS transistor **500** is in the “off” state (i.e., non-conducting), and negligible current flows between the source/body terminal **504** and the drain/N-iso terminal **508**.

[0061] FIGS. 6 and 7 are schematic circuit diagrams each illustrating variations on the same multiplexers **300**, **400** illustrated in FIGS. 3 and 4, but with parasitic PNP (positive-negative-positive) transistors included. Specifically, FIG. 6 illustrates a center switch **600** that corresponds with the center switch of the multiplexer **300** illustrated in FIG. 3. FIG. 7 illustrates a center switch **700** that corresponds with the center switch of the multiplexer **400** illustrated in FIG. 4. In both configurations, the reverse biasing of the lower-voltage FET turns on the source/body to drain/N-iso junction. Excess current flows as shown by the arrows.

[0062] In FIG. 6, the center switch **600** includes the second source resistor **304b**, the second left NMOS transistor **306b**, the second right NMOS transistor **308b**, and a parasitic PNP transistor **612**. In FIG. 7, the center switch **700** includes the second source resistor **404b**, the second left NMOS

transistor **406b**, the second right NMOS transistor **408b**, and a parasitic PNP transistor **712**. In both center switches **600**, **700**, the reverse biasing of the lower-voltage FET turns on the source/body to drain/N-iso junction. This causes excess current to flow as shown in FIGS. **6** and **7**.

[0063] FIGS. **8-10** are schematic circuit diagrams each illustrating a switch that includes parking to prevent parasitic PNP leakage. FIG. **8** is a schematic circuit diagram of an exemplary switch **800** that could be implemented as a center switch in the multiplexers **300**, **400** described herein. FIG. **9** is a schematic circuit diagram of an exemplary switch **900** that could be implemented as a center switch in the multiplexers **300**, **400** described herein. FIG. **10** is a schematic circuit diagram of an exemplary switch **1000** that could be implemented as a center switch in the multiplexers **300**, **400** described herein.

[0064] In FIG. **8**, the switch **800** includes a source resistor **804**, a left NMOS transistor **806**, and a right NMOS transistor **808**. The switch **800** additionally includes a common-source node **816** and a midpoint NMOS transistor **814** in electrical communication with the common-source node **816**. The common-source node **816** is a point in the switch **800** where multiple components or signals share a common connection. Specifically, the left NMOS transistor **806** and the right NMOS transistor **808** share a common connection at the common-source node **816**. The switch **800** includes a parasitic PNP transistor like the embodiments illustrated in FIGS. **6** and **7**, but the switch **800** prevents the parasitic PNP transistor from turning on.

[0065] The switch **800** prevents the parasitic PNP transistor from turning on because the switch **800** includes the midpoint NMOS transistor **814** in electrical communication with the common-source node **816**. Rather than allowing the common-source node **816** to float when the switch **800** is off, the parasitic PNP transistor is prevented from turning on by “parking” the midpoint NMOS transistor **814** to a low voltage. In the implementation illustrated in FIG. **8**, the common-source node **816** is parked at 20 V. The midpoint NMOS transistor **814** is a switch that connects the park voltage input 20 V to the common-source node **816**. The 25 V on the gate of the midpoint NMOS transistor **814** supplies the 5 V V_{gs} to turn on the midpoint NMOS transistor **814**. By pulling the midpoint NMOS transistor **814** low, the source/body to drain/N-iso junctions of the left NMOS transistor **806** and the right NMOS transistor **808** are kept reverse-biased. This reduces or eliminates excess current flow.

[0066] In FIG. **8**, the switch **800** is off, and the common-source node **816** is parked at 20 V. In this case, the switch **800** includes two input voltages, including the 30 V input voltage and the 20 V park voltage input. The switch includes one output voltage, namely the output voltage of 40 V from the right NMOS transistor **808**. The drains of the left NMOS transistor **806** and the right NMOS transistor **808** are each withstanding high voltage.

[0067] In FIG. **9**, the switch **900** includes a source resistor **904**, a left NMOS transistor **906**, and a right NMOS transistor **908**. The switch **900** additionally includes a common-source node **916** and a midpoint NMOS transistor **914** in electrical communication with the common-source node **916**. In the switch **900**, the left NMOS transistor **906** and the right NMOS transistor **908** share a common connection at the common-source node **916**. Notably, the switch **900** includes a parasitic PNP transistor (not shown), but the parasitic PNP is not on.

[0068] Like in the switch **800** illustrated in FIG. **8**, the switch **900** prevents the parasitic PNP transistor from turning on because the switch **900** includes the midpoint NMOS transistor **914** in electrical communication with the common-source node **916**. Rather than allowing the common-source node **916** to float when the switch **900** is off, the parasitic PNP transistor is prevented from turning on by “parking” the midpoint NMOS transistor **914** to a low voltage. In the implementation illustrated in FIG. **9**, the midpoint NMOS transistor **914** is parked at 20 V. By pulling the midpoint NMOS transistor **914** low, the source/body to drain/N-iso junctions of the left NMOS transistor **906** and the right NMOS transistor **908** are kept reverse-biased. This reduces or eliminates excess current flow.

[0069] In FIG. **9**, the switch **900** is off, and the midpoint NMOS transistor **914** is parked at 20 V. In

this case, the switch **900** receives the input voltage of 30 V and the park voltage (also an input) of 20 V. The switch **900** outputs a multiplexer output voltage of 20 V from the right NMOS transistor **908**. The drain of the left NMOS transistor **906** is withstanding high voltage.

[0070] In the switch **800** illustrated in FIG. 8, and further in the switch **900** illustrated in FIG. 9, the park voltage serves as an input. When the multiplexer switch is in an OFF state, the park transistor (i.e., the midpoint NMOS transistor **814** or the midpoint NMOS transistor **914**) turns on. This pulls the midpoint (i.e., the common-source node **816** or common-source node **916**) down to the park voltage input. In order for the parking function to work, the park voltage must be less than or equal to the lowest multiplexer input voltage. FIG. 8 illustrates an example wherein the park voltage (20 V) is less than the input voltage (30 V) and the multiplexer output (40 V). In this case, both the left NMOS transistor **806** and the right NMOS transistor **808** withstand high voltage from drain to source. FIG. 9 illustrates a different case, wherein the park voltage (20 V) is below the switch input voltage (30 V) but equal to the multiplexer output voltage (20 V). In this case, only the left NMOS transistor **906** withstands high drain-to-source voltage, because the multiplexer output (20 V) is equivalent to the park voltage (20 V).

[0071] In FIG. 10, the switch **1000** includes a source resistor **1004**, a left NMOS transistor **1006**, and a right NMOS transistor **1008**. In contrast with the switches **800**, **900** illustrated in FIGS. 8 and 9, the switch **1000** is turned on and additionally includes a gate **1002**. The switch **1000** includes a common-source node **1016** and a midpoint NMOS transistor **1014** in electrical communication with the common-source node **1016**. In the switch **1000**, the left NMOS transistor **1006** and the right NMOS transistor **1008** share a common connection at the common-source node **1016**.

[0072] In FIG. 10, the switch **1000** is on, and the midpoint NMOS transistor **1014** is not parked. The drain of the midpoint NMOS transistor **1014** is withstanding high voltage. As described herein, a drain withstands high voltage when the drain-to-source voltage is higher than the maximum allowed gate-to-source voltage. It is not necessary that the switch **1000** actually withstands high voltage, even though the switch **1000** is designed to withstand high voltage. In the example illustrated in FIG. 10, the park voltage is 20 V, the multiplexer output voltage is 30 V, and thus the midpoint NMOS transistor **1014** withstands high voltage. In an alternative exemplary implementation, if the park voltage was 20 V and the multiplexer output voltage was 25 V, then the midpoint NMOS transistor **1014** would not withstand high voltage, but the functionality of the switch **1000** would be the same.

EXAMPLES

[0073] The following examples pertain to further embodiments.

[0074] Example 1 is a switch. The switch includes a source resistor; a first transistor, wherein the first transistor is a metal-oxide-semiconductor field-effect (MOSFET) transistor; a second transistor, wherein the second transistor is a MOSFET transistor; and a midpoint transistor, wherein the midpoint transistor is a MOSFET transistor, and wherein the midpoint transistor is located in between the first transistor and the second transistor. The midpoint transistor is maintained at a low voltage in response to the switch being in a non-conductive state, and the low voltage is less than or equal to an output voltage of the switch.

[0075] Example 2 is a switch as in Example 1, wherein the first transistor comprises: a first gate; a first source terminal; and a first drain/N-iso terminal; and wherein the second transistor comprises: a second gate; a second source terminal; and a second drain/N-iso terminal.

[0076] Example 3 is a switch as in any of Examples 1-2, wherein, in response to the midpoint transistor maintaining the low voltage: a first junction from the first source terminal to the first drain/N-iso terminal of the first transistor is reverse biased; and a second junction from the second source terminal to the second drain/N-iso terminal of the second transistor is reverse biased.

[0077] Example 4 is a switch as in any of Examples 1-3, wherein the midpoint transistor is maintained at the low voltage to reduce excess current flow in the switch.

[0078] Example 5 is a switch as in any of Examples 1-4, wherein the switch comprises a

multiplexer voltage output that is in direct electrical communication with the second transistor; and wherein the switch comprises two voltage inputs comprising: a first voltage input in direct electrical communication with the first transistor; and a midpoint voltage input in direct electrical communication with the midpoint transistor.

[0079] Example 6 is a switch as in any of Examples 1-5, wherein the midpoint voltage input comprises a midpoint voltage level; wherein the multiplexer output comprises a multiplexer voltage level; and wherein, in response to the midpoint transistor being maintained at the low voltage, the midpoint voltage level is less than the multiplexer voltage level.

[0080] Example 7 is a switch as in any of Examples 1-6, wherein the midpoint voltage input comprises a midpoint voltage level; wherein the multiplexer output comprises a multiplexer voltage level; and wherein, in response to the midpoint transistor being maintained at the low voltage, the midpoint voltage level is substantially equal to the multiplexer voltage level.

[0081] Example 8 is a switch as in any of Examples 1-7, wherein the midpoint transistor is not maintained at the low voltage in response to the switch being in a conductive state.

[0082] Example 9 is a switch as in any of Examples 1-8, wherein, in response to the switch being in the conductive state, the midpoint transistor withstands a high voltage drop, and wherein the high voltage drop is less than or equal to the output voltage of the switch.

[0083] Example 10 is a switch as in any of Examples 1-9, wherein, in response to the midpoint transistor being maintained at the low voltage, each of the first transistor and the second transistor withstands a high voltage drop, and wherein the high voltage drop is less than or equal to the output voltage of the switch.

[0084] Example 11 is a switch as in any of Examples 1-10, wherein, in response to the midpoint transistor being maintained at the low voltage, one of the first transistor or the second transistor withstands a high voltage drop, and wherein the high voltage drop is less than or equal to the output voltage of the switch.

[0085] Example 12 is a switch as in any of Examples 1-11, wherein at least one of the first transistor, the second transistor, or the midpoint transistor is an N-channel metal-oxide semiconductor (NMOS) transistor.

[0086] Example 13 is a switch as in any of Examples 1-12, wherein at least one of the first transistor, the second transistor, or the midpoint transistor is a double diffusion metal oxide semiconductor field-effect (DMOS FET) transistor.

[0087] Example 14 is a switch as in any of Examples 1-13, wherein maintaining the midpoint transistor at the low voltage substantially eliminates a parasitic positive-negative-positive (PNP) transistor occurring within either of the first transistor or the second transistor.

[0088] Example 15 is a switch as in any of Examples 1-14, wherein the switch is one of a plurality of switches in a multiplexer.

[0089] Example 16 is a switch as in any of Examples 1-15, wherein the plurality of switches of the multiplexer comprises: a first switch receiving a first input voltage; a second switch receiving a second input voltage; and a third switch receiving a third input voltage; wherein the first input voltage is greater than either of the second input voltage or the third input voltage; and wherein the second input voltage is greater than the third input voltage.

[0090] Example 17 is a switch as in any of Examples 1-16, wherein, in response to the first switch being in a conductive state and each of the second switch and the third switch being in the non-conductive state, an output voltage of the multiplexer is substantially equal to the first input voltage.

[0091] Example 18 is a switch as in any of Examples 1-17, wherein, in response to the third switch being in a conductive state and each of the first switch and the second switch being in the non-conductive state, an output voltage of the multiplexer is substantially equal to the third input voltage.

[0092] Example 19 is a switch as in any of Examples 1-18, wherein the multiplexer is a component

of a battery management system.

[0093] Example 20 is a switch as in any of Examples 1-19, wherein the battery management system is configured to manage a plurality of batteries, and wherein a quantity of the plurality of batteries is evenly divisible by a quantity of the plurality of switches such that each of the plurality of switches is in electrical communication with a same number of batteries of the plurality of batteries.

[0094] Example 21 is a multiplexer. The multiplexer comprises a switch includes any of the characteristics of Examples 1-20.

[0095] Example 22 is a battery management system. The battery management system includes a switch comprising any of the characteristics of Examples 1-20.

[0096] The foregoing description has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise form disclosed. Many modifications and variations are possible considering the above teaching. Further, it should be noted that any or all the alternate implementations may be used in any combination desired to form additional hybrid implementations of the disclosure.

[0097] Further, although specific implementations of the disclosure have been described and illustrated, the disclosure is not to be limited to the specific forms or arrangements of parts so described and illustrated. The scope of the disclosure is to be defined by the claims appended hereto, if any, any future claims submitted here and in different applications, and their equivalents.

[0098] It is to be understood that any features of the above-described arrangements, examples, and embodiments may be combined in a single embodiment comprising a combination of features taken from any of the disclosed arrangements, examples, and embodiments.

[0099] It will be appreciated that various features disclosed herein provide significant advantages and advancements in the art. The following claims are exemplary of some of those features.

[0100] In the foregoing Detailed Description of the Disclosure, various features of the disclosure are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed disclosure requires more features than are expressly recited in each claim. Rather, inventive aspects lie in less than all features of a single foregoing disclosed embodiment.

[0101] It is to be understood that the above-described arrangements are only illustrative of the application of the principles of the disclosure. Numerous modifications and alternative arrangements may be devised by those skilled in the art without departing from the spirit and scope of the disclosure and the appended claims are intended to cover such modifications and arrangements. Thus, while the disclosure has been shown in the drawings and described above with particularity and detail, it will be apparent to those of ordinary skill in the art that numerous modifications, including, but not limited to, variations in size, materials, shape, form, function and manner of operation, assembly and use may be made without departing from the principles and concepts set forth herein.

[0102] The foregoing description has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise form disclosed. Many modifications and variations are possible considering the above teaching. Further, it should be noted that any or all the alternate implementations may be used in any combination desired to form additional hybrid implementations of the disclosure.

[0103] Further, although specific implementations of the disclosure have been described and illustrated, the disclosure is not to be limited to the specific forms or arrangements of parts so described and illustrated. The scope of the disclosure is to be defined by the claims appended hereto, any future claims submitted here and in different applications, and their equivalents.

Claims

- 1.** A switch comprising: a source resistor; a first transistor, wherein the first transistor is a metal-oxide-semiconductor field-effect (MOSFET) transistor; a second transistor, wherein the second transistor is a MOSFET transistor; and a midpoint transistor, wherein the midpoint transistor is a MOSFET transistor, and wherein the midpoint transistor is located in between the first transistor and the second transistor; wherein the midpoint transistor is maintained at a low voltage in response to the switch being in a non-conductive state; and wherein the low voltage is less than or equal to an output voltage of the switch.
- 2.** The switch of claim 1, wherein the first transistor comprises: a first gate; a first source terminal; and a first drain/N-iso terminal; and wherein the second transistor comprises: a second gate; a second source terminal; and a second drain/N-iso terminal.
- 3.** The switch of claim 2, wherein, in response to the midpoint transistor maintaining the low voltage: a first junction from the first source terminal to the first drain/N-iso terminal of the first transistor is reverse biased; and a second junction from the second source terminal to the second drain/N-iso terminal of the second transistor is reverse biased.
- 4.** The switch of claim 1, wherein the midpoint transistor is maintained at the low voltage to reduce excess current flow in the switch.
- 5.** The switch of claim 1, wherein the switch comprises a multiplexer voltage output that is in direct electrical communication with the second transistor; and wherein the switch comprises two voltage inputs comprising: a first voltage input in direct electrical communication with the first transistor; and a midpoint voltage input in direct electrical communication with the midpoint transistor.
- 6.** The switch of claim 5, wherein the midpoint voltage input comprises a midpoint voltage level; wherein the multiplexer output comprises a multiplexer voltage level; and wherein, in response to the midpoint transistor being maintained at the low voltage, the midpoint voltage level is less than the multiplexer voltage level.
- 7.** The switch of claim 5, wherein the midpoint voltage input comprises a midpoint voltage level; wherein the multiplexer output comprises a multiplexer voltage level; and wherein, in response to the midpoint transistor being maintained at the low voltage, the midpoint voltage level is substantially equal to the multiplexer voltage level.
- 8.** The switch of claim 1, wherein the midpoint transistor is not maintained at the low voltage in response to the switch being in a conductive state.
- 9.** The switch of claim 8, wherein, in response to the switch being in the conductive state, the midpoint transistor withstands a high voltage drop, and wherein the high voltage drop is less than or equal to the output voltage of the switch.
- 10.** The switch of claim 1, wherein, in response to the midpoint transistor being maintained at the low voltage, each of the first transistor and the second transistor withstands a high voltage drop, and wherein the high voltage drop is less than or equal to the output voltage of the switch.
- 11.** The switch of claim 1, wherein, in response to the midpoint transistor being maintained at the low voltage, one of the first transistor or the second transistor withstands a high voltage drop, and wherein the high voltage drop is less than or equal to the output voltage of the switch.
- 12.** The switch of claim 1, wherein at least one of the first transistor, the second transistor, or the midpoint transistor is an N-channel metal-oxide semiconductor (NMOS) transistor.
- 13.** The switch of claim 1, wherein at least one of the first transistor, the second transistor, or the midpoint transistor is a double diffusion metal oxide semiconductor field-effect (DMOS FET) transistor.
- 14.** The switch of claim 1, wherein maintaining the midpoint transistor at the low voltage prevents activation of a parasitic positive-negative-positive (PNP) transistor within either of the first transistor or the second transistor.
- 15.** The switch of claim 1, wherein the switch is one of a plurality of switches in a multiplexer.
- 16.** The switch of claim 15, wherein the plurality of switches of the multiplexer comprises: a first

switch receiving a first input voltage; a second switch receiving a second input voltage; and a third switch receiving a third input voltage; wherein the first input voltage is greater than either of the second input voltage or the third input voltage; and wherein the second input voltage is greater than the third input voltage.

17. The switch of claim 16, wherein, in response to the first switch being in a conductive state and each of the second switch and the third switch being in the non-conductive state, an output voltage of the multiplexer is substantially equal to the first input voltage.

18. The switch of claim 16, wherein, in response to the third switch being in a conductive state and each of the first switch and the second switch being in the non-conductive state, an output voltage of the multiplexer is substantially equal to the third input voltage.

19. The switch of claim 16, wherein the multiplexer is a component of a battery management system.

20. The switch of claim 19, wherein the battery management system is configured to manage a plurality of batteries, and wherein a quantity of the plurality of batteries is evenly divisible by a quantity of the plurality of switches such that each of the plurality of switches is in electrical communication with a same number of batteries of the plurality of batteries.
